KOSZUL AND ALTERNATING CECH COMPLEXES DETECT QUASI-ISOMORPHISMS MODULO I FOR MAPS IDENTIFYING R/I AND S/IS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- A map of complexes K -> K tensor S over a flat ring map is a quasi-isomorphism as soon as the homology groups of K are I-power torsion and the map induces an isomorphism R/I -> S/IS on a neighbourhood. invoked at proof lines 24386-24388. Reduces both claims to the torsion property of the homology.
- The homology groups of the Koszul complex K(R, f_1, ..., f_r) are I-power torsion when V(f_1, ..., f_r) = V(I). invoked at proof lines 24388-24389. Gives the torsion property for the Koszul case.
- The homology groups of the extended alternating Cech complex on f_1, ..., f_r are I-power torsion. invoked at proof lines 24389-24391. Gives the torsion property for the Cech case.

Source reference. Original lemma 05E9: sources/stacks-more-algebra/pinned-original-source.tex, source lines 24264–24264. Cited at proof lines 24386-24387 for the reduction to I-power torsion homology; the conclusion is used inside the counted span.

Source reference. Original lemma 0663: sources/stacks-more-algebra/pinned-original-source.tex, source lines 6660–6660. Cited at proof lines 24388-24389 for the torsion property in the Koszul case; the conclusion is used inside the counted span.

Source reference. Original lemma 0G6K: sources/stacks-more-algebra/pinned-original-source.tex, source lines 7066–7066. Cited at proof lines 24389-24391 for the torsion property in the Cech case; the conclusion is used inside the counted span.

Lemma 0.2. *Assume $\varphi : R \to S$ is a flat ring map and $I \subset R$ is a finitely generated ideal such that $R/I \to S/IS$ is an isomorphism. For any $f_1, \ldots, f_r \in R$ such that $V(f_1, \ldots, f_r) = V(I)$*

(1) *the map of Koszul complexes $K(R, f_1, \ldots, f_r) \to K(S, f_1, \ldots, f_r)$ is a quasi-isomorphism, and*

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(2) *The map of extended alternating Čech complexes*

$$\begin{array}{ccccccc} R \to \prod_{i_0} R_{f_{i_0}} & \to & \prod_{i_0<i_1} R_{f_{i_0}f_{i_1}} & \to \ldots \to & R_{f_1\ldots f_r} \\ \downarrow & & \downarrow & & \\ S \to \prod_{i_0} S_{f_{i_0}} & \to & \prod_{i_0<i_1} S_{f_{i_0}f_{i_1}} & \to \ldots \to & S_{f_1\ldots f_r} \end{array}$$

is a quasi-isomorphism.

Proof. In both cases we have a complex $K_\bullet$ of R modules and we want to show that $K_\bullet \to K_\bullet \otimes_R S$ is a quasi-isomorphism. By Lemma 05E9 and the flatness of $R \to S$ this will hold as soon as all homology groups of K are I-power torsion. This is true for the Koszul complex by Lemma 0663 and for the extended alternating Čech complex by Lemma 0G6K. □